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DI TRENTO

HTML: the language for the web

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What is HTML?

Hypertext Markup Language is the standard markup language for creating Web pages

...that is the language used to create hypertext documents that are portable from one platform to another.

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Hypertext

Hypertext is text displayed on a computer display or other electronic devices with references (**hyperlinks**) to other text that the reader can immediately access.

Hypertext documents are interconnected by hyperlinks, which are typically activated by user actions, such as a mouse click, keypress set or by touching the screen.



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Where is HTML defined?

- HTML 2.0 published as standard with in RFC 1866

[\[RFC Home\]](#) [\[TEXT\]](#) [\[PDF\]](#) [\[HTML\]](#) [\[Tracker\]](#) [\[IPR\]](#) [\[Info page\]](#)

Obsoleted by: [2854](#) HISTORI

Network Working Group T. Berners-Lee
Request for Comments: 1866 MIT/W3C
Category: Standards Track D. Connolly
November 1995

Hypertext Markup Language - 2.0

Status of this Memo

This document specifies an Internet standards track protocol for the Internet community, and requests discussion and suggestions for improvements. Please refer to the current edition of the "Internet Official Protocol Standards" (STD 1) for the standardization state and status of this protocol. Distribution of this memo is unlimited

Abstract

The Hypertext Markup Language (HTML) is a simple markup language use to create hypertext documents that are platform independent. HTML documents are SGML documents with generic semantics that are appropriate for representing information from a wide range of domains. HTML markup can represent hypertext news, mail, documentation, and hypermedia; menus of options; database query results; simple structured documents with in-lined graphics; and hypertext views of existing bodies of information.

- HTML standards were initially developed under the authority of the World Wide Web Consortium (W3C)
<http://www.w3c.org>

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HTML versions

Version	Date	Notes
HTML 2.0	1995	First official version
HTML 3.2	1997	
HTML 4.0 (4.01)	1999	Add table, CSS, JS
XHTML 1.0	2000	XML version of HTML 4
DHTML		HTML 4 +JS + CSS + DOM
XHTML 2.0		Abandoned in 2009
HTML 5 (5.2)	2014 (2017)	(2008 first draft, 2021 last version)
XHTML 5.1		XML version of HTML 5

- Initially maintained by W3C
- Since 2012 also defined by WHATWG (Web Hypertext Application Technology Working Group)
- Since 2019 maintained only by WHATWG

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HTML Principles

- Two original principles:
 - Graceful degradation of presentation
 - A more philosophical rule: “[...] HTML should convey the structure of a hypertext document, but not the details of its presentation.”
- After a while HTML has evolved from a sematic language into a presentation language

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Example of graceful degradation

Let's consider this fragment of language written in HTML 5:

```
<audio src="teenage_dream.ogg" autoplay controls loop>
  <a href="teenage_dream.ogg">listen</a>
</audio>
```

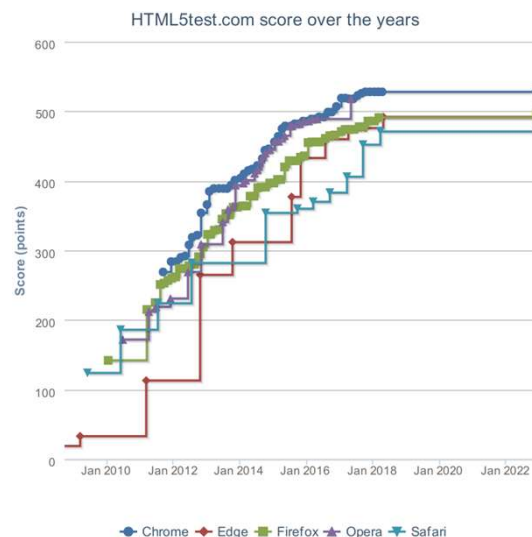


What if my browser just supports HTML 4?

```
<audio src="teenage_dream.ogg" autoplay controls loop>
  <a href="teenage_dream.ogg">listen</a>
</audio>
```

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HTML5 over the years



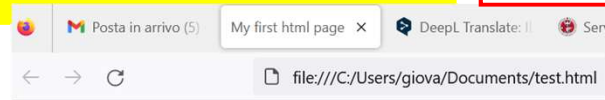
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Structure of an HTML doc

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>My first html page</title>
  </head>
  <body>
    <p>Welcome in my first page</p>
  </body>
</html>
```

Note that:

- Indentation is irrelevant, everything will be mapped onto a single space
- No case sensitivity (lowercase is preferred)
- HTML works without a server (try on your laptop!)



Welcome in my first page

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DOCTYPE Declaration

```
<!DOCTYPE html>
<html>
...
</html>
```

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
```

Today (simplified in HTML 5.0):
<!DOCTYPE html>

<html> is the root of a html page

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The HTML DTDs!

<https://www.w3.org/TR/REC-html40/sgml/dtd.html>

<https://www.w3.org/TR/html401/sgml/dtd.html>

https://www.w3.org/TR/xhtml1/dtds.html#a_dtd_XHTML-1.0-Strict

There is no HTML5 DTD!

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<head>

```
<head>
  <meta charset="utf-8">
  <title>My first html page</title>
</head>
```

HEAD: contains information (metadata) about the document:

- Meta tags: used to describe the content (character set, author, keywords for search engines, description...)
- Title of the page (appearing at the top of the browser window)
- Statements or references to JavaScript and StyleSheets code

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Character Set

- To display an HTML page correctly, a web browser must know which character set to use.
- It describes the mapping between chars and integer numbers:
 - **ASCII** was the first character set. 128 different alphanumeric characters: {numbers (0-9), English letters (A-Z), some special characters e.g. ! \$ + - () @ < >}
 - **ISO-8859-1**: default character set for HTML 4. This character set supported 256 different character codes
 - **ANSI** (Windows-1252): the original Windows character set. Add 32 extra characters to ISO-8859-1
 - **UTF-8** (Unicode) was supported by HTML4, is the default character set for HTML5.

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Entities

&entity_name

or

&#entity_number

Result	Description	Entity Name	Entity Number
	non-breaking space	 	
<	less than	<	<
>	greater than	>	>
&	ampersand	&	&
"	double quotation mark	"	"
'	single quotation mark (apostrophe)	'	'
¢	cent	¢	¢
£	pound	£	£
¥	yen	¥	¥
€	euro	€	€
©	copyright	©	©
®	registered trademark	®	®

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URL Encoding

Character	From Windows-1252	From UTF-8
space	%20	%20
!	%21	%21
"	%22	%22
#	%23	%23
\$	%24	%24
%	%25	%25
&	%26	%26
'	%27	%27
(	%28	%28
)	%29	%29

See also https://www.w3schools.com/tags/ref_urlencode.asp

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<body>

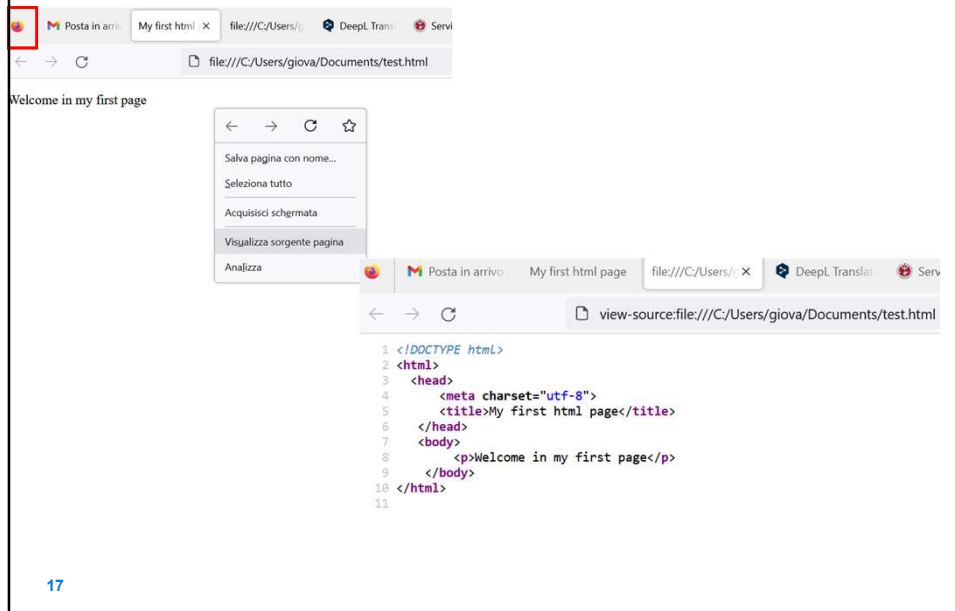
```
<body>
  <p>Welcome in my first page</p>
</body>
```

<body>: contains the actual content of the document.

This is the part that will be displayed in the browser window.

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View source



HTML Tags

Learning HTML essentially means learning a set of tags

Remember that HTML is tolerant! Some examples:

- Empty element: `<br>` does not strictly require a closing tag, in principle it would be better to use `<br></br>` or `<br/>`

Is the Ending Slash Optional?

HTML5: the slash is optional.

HTML4: the slash is technically invalid.

However, it's accepted by W3C's HTML validator.

XHTML: The slash is REQUIRED.

- Nested tags:

**** The winner is: *<I>* Sofia Goggia! **** Congratulations! *</I>*

Attributes

- Attributes provide additional information about elements
- Attributes are always specified in the start tag
- Attributes usually come in name/value pairs like:
name="value"
- You can add multiple attributes within a single tag
- Some attributes only have name (no associated value)
- Values are limited to 1024 characters in length
- Attributes values are within single or double quotes.

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Basic formatting tags

- Headings **<h1>...<h6>**
- Emphasis **, , , <i>**
- Subscript, superscript **<sub><sup>**

```
<!DOCTYPE html>
<html>
<body>
<h1>this is H1</h1>
<h2>this is H2</h2>
<h3>this is H3</h3>
<h4>this is H4</h4>
this is normal text<br>
<b>this is bold text</b><br>
<em>this is emphasized text</em><br>
<strong>this is strong text</strong>
<br>
<cite>this is a citation</cite><br>
</body>
</html>
```

this is H1

this is H2

this is H3

this is H4

this is normal text

this is bold text

this is emphasized text

this is strong text

this is a citation

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Basic formatting tags

- Lists ****, ****, ****, **<dl>**, **<dt>**, **<dd>**

```
<ol>
  <li>...</li>
  <li>...</li>
</ol>
```

OL - Ordered List

1. First item
2. Second item
3. Third item

UL – Unordered List

- First item
- Second item
- Third item

DL – Description List

First item
Description of 1st item
Second item
Description of 2nd item

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Basic formatting: blocks level vs inline

- Each HTML element has a default **display value**: block or inline:
 - **Block-level elements** always start on a new line, and the browsers automatically add some space (a margin) before and after the element. Block-level elements always take up the full width available (stretches out to the left and right as far as it can).
 - **Inline elements** do not start on a new line and only take up as much width as necessary.

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Basic formatting: blocks

- Paragraphs and blocks: `<p>`, `<div>`, `<span>`
- `<p>` and `<div>` are block elements with *semantic* difference:
 - `<div>` describes a container of data (with graphic separation)
 - `<p>` describes a paragraph of content (with graphic separation). It cannot contain block-level elements (P, DIV) but can contain SPAN, but HTML is tolerant...
- `<span>` is a logical, in-line grouping without graphic evidence. It's mostly used for a small chunk of HTML inside a line

Don't mark up your document based on how it should look - mark it up based on what it is.

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Image and figure

According to W3C:

"The figure element represents a unit of content, optionally with a caption, that is self-contained, that is typically referenced as a single unit from the main flow of the document, and that can be moved away from the main flow of the document without affecting the document's meaning."

- Image vs Figure:
 - Image is used to embed the image in a HTML document
 - Figure is used to semantically organize the content of an image in the HTML document

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Semantic difference

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Using the image tag</title>
</head>
<body>
  <p>
    I will be attaching an image of a dog here below.
  </p>
  <div>
    
    <p>
      Here is an image of a dog.
    </p>
  </div>
</body>
</html>
```

I will be attaching an image of a dog here below.



Here is an image of a dog.

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Semantic difference

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Using the figure tag</title>
</head>
<body>
  <p>
    I will be attaching an image of a dog here below.
  </p>
  <figure>
    
    <figcaption>Here is a image of a dog.</figcaption>
  </figure>
</body>
</html>
```

I will be attaching an image of a dog here below.



Here is a image of a dog.

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File paths

- A file path describes the location of a file in a website's folder structure
- They are used when we need to link to external files like:
 - Web pages
 - Images
 - Style sheets
 - JavaScripts

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File paths

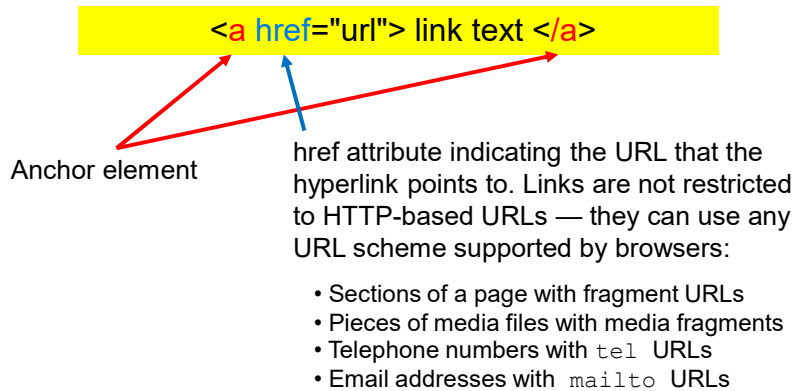
<code></code>	Absolute Path
<code></code>	picture.jpg is located in the same folder as the current page
<code></code>	picture.jpg is located in the images folder inside the current folder
<code></code>	picture.jpg is located in the images folder at the root of the current web app
<code></code>	picture.jpg is located in the folder one level up from the current folder

It is best practice to use relative file paths (if possible)

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Hyperlinks

- You can click on a link and jump to another document



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Hyperlinks

```
<p>You can reach me at:</p>
<ul>
  <li><a href="https://mywebsite_gv.com"> My website</a></li>
  <li><a href="mailto:giovanna.varni@unitn.it">My email</a></li>
  <li><a href="tel:+123456789"> My phone</a></li>
</ul>
```

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Internal Hyperlinks

- HTML internal link is linked within the same web page
- This link can be an absolute path or relative path

```

<!DOCTYPE html>
<html>
<head></head>
<body>
<a href="#lesson1">Lesson.1</a><br />
<a href="#lesson2">Lesson.2</a><br />
<a href="#lesson3">Lesson.3</a><br />
<br />

<h1 id="lesson1">Introduction of Lesson.1</h1>
<p>This is sub topic.1</p>
<p>This is sub topic.2</p>
<br />

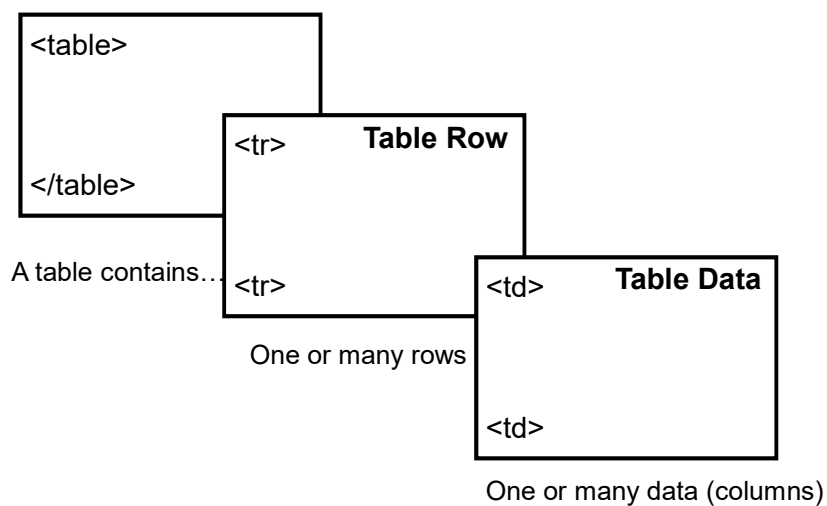
<h1 id="lesson2">Introduction of Lesson.2</h1>
<p>This is sub topic.1</p>
<p>This is sub topic.2</p>
<br />
...

```

GO TO Lesson 1

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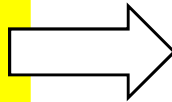
Tables



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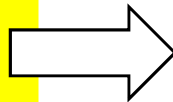
Tables

```
<table>
  <tr>
    <td> Cell 1 </td>
    <td> Cell 2 </td>
  </tr>
</table>
```



Cell 1	Cell 2
--------	--------

```
<table>
  <tr>
    <td> Cell 1 </td>
  </tr>
  <tr>
    <td> Cell 2 </td>
  </tr>
</table>
```

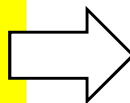


Cell 1
Cell 2

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Tables

```
<table>
  <tr>
    <td> Cell 1 </td>
    <td> Cell 2 </td>
  </tr>
  <tr>
    <td colspan=2> Cell 3 </td>
  </tr>
</table>
```



Cell 1	Cell 2
Cell 3	

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Tables

<h2>Cell that spans two columns</h2>

```
<table style="width:100%">
  <tr>
    <th>Name</th>
    <th colspan="2">Telephone</th>
  </tr>
  <tr>
    <td>Bill Gates</td>
    <td>555 77854</td>
    <td>555 77855</td>
  </tr>
  <tr>
    <td>Donald Duck</td>
    <td>945 2626</td>
    <td>0471 23706</td>
  </tr>
</table>
```

Cell that spans two columns

Name	Telephone	
Bill Gates	555 77854	555 77855
Donald Duck	945 2626	0471 23706

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Practicing @home

Create a webpage that looks like this:

1,1	2,1	3,1	4,1	5,1	6,1	7,1	8,1	9,1	10,1
2,2	4,2	5,2	6,2	7,2	8,2	9,2	10,2		
	4,3	5,3	6,3	7,3	8,3	9,3	10,3		
2,4						7,4	8,4	9,4	10,4
2,5						7,5	8,5	9,5	10,5
1,2	2,6					7,6	8,6	9,6	10,6
2,7	4,7								
2,8	4,8								
2,9	4,9								
2,10	4,10								

Suggestion: bgcolor="red"

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Form

- A form enables to collect (in an interactive way) information from the user
- This information is then passed to a back-end application, such as CGI, PHP script, and so on that will handle the data according to the logic defined within the back-end application
- How do we can collect / submit information?
Through text input field, radio button, checkbox and so on

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Form and its attributes

```
<form action="Script URL" method="GET|POST" >
...
form elements
...
</form>
```

- Form attributes:
 - action: the URL that identifies the resource processing the form submission
 - method: the HTTP method to submit the form with
 - autocomplete: specifies whether a form should have autocomplete on or off. When autocomplete is on, the browser automatically complete values based on values that the user has entered before
 - enctype: specifies the MIME type used to send data to the server. It can be used only when the method is POST.

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Form and its elements

```
<form action="Script URL" method="GET|POST" >
```

```
...
```

```
form elements
```

```
...
```

```
</form>
```

<input>	<fieldset>
<label>	<legend>
<textarea>	<datalist>
<button>	<option>
<select>	...

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<input>

Some input types:

<input type="text">	<input type="button">
<input type="password">	<input type="submit">
<input type="radio">	<input type="reset">
<input type="email">	<input type="file">
<input type="checkbox">	<input type="hidden">

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<input>: example

```
<form action="/action_page.php" method="POST">
  <label for="fname">Enter your first name:</label>
  <input type="text" id="fname" name="fname"><br>
<br>
  <label for="lname">Enter your last name:</label>
  <input type="text" id="lname" name="lname"><br>
<br>
  <label for="email">Enter your email:</label>
  <input type="email" id="email" name="email"><br>
<br>
  <input type="submit" value="Submit">
</form>
```

Enter your first name:

Enter your last name:

Enter your email:

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hidden

- `<input type="hidden">` is used by developers to send to a server additional data which are not set in the form by users

```
<form>
  <div>
    <label for="title">Post title:</label>
    <input type="text" id="title" name="title" value="My post" />
  </div>
  <div>
    <label for="content">Post content:</label>
    <input type="text" id="content" name="content" value="Hello" />
  </div>
  <div>
    <button type="submit">Update post</button>
  </div>
  <input type="hidden" id="hidden" name="my_hidden" value="Nobody can see me" />
</form>
```

file:///C:/Users/giova/Desktop/test.html?title=My+post&content=Hello&my_hidden=Nobody+can+see+mee

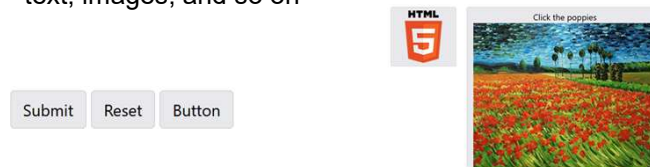
Post title:

Post content:

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<button> or <input>?

- Both <button> and <input> elements come in 3 types:
 - type="submit": submits all input elements in a form to the server
 - type="reset": resets all input elements in a form to its original values
 - type="button": standard button that usually invokes a JavaScript function
- Differences:
 - <input> buttons can only have text, whereas <button> can contain text, images, and so on



Source: <https://www.dofactory.com/html/buttons>

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How many buttons?

- In HTML 4 at most 2 and only one interacts with the server (the submit button)
- In HTML 5 as many as you want!

The HTML5's "formaction" attribute:

- overrides the form action attribute for the element <button> and <input> WITH type="submit"
- specifies where to send the data when a form is submitted
- gives us the ability to define as many buttons as we need and each of button can point to a different url

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formation

```
<!DOCTYPE html>
<html>
<body>

<h1>The button formation attribute</h1>

<form method="get">
  <label for="fname">First name:</label>
  <input type="text" id="fname" name="fname"><br><br>
  <label for="lname">Last name:</label>
  <input type="text" id="lname" name="lname"><br><br>
  <button type="submit" formation= "/action_page.php">Submit</button>
  <button type="submit" formation="/action_page2.php">Submit to another
page</button>
</form>

</body>
</html>
```

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Practicing @home



Foto di [Dan Gold](#) su [Unsplash](#)

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Let's install a (Apache) web server

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Let's install a Web Server

We will, step by step:

1. Install Apache (with a series of extra tools: DB and languages – Perl and PHP)
2. Customize it (e.g. by changing port if needed)
3. Customize its response (static files)

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XAMPP

<https://www.apachefriends.org/download.html>

What does **XAMPP** means?

Generalization of **WAMPP**, **LAMPP**, **MAMPP**

W = Windows, L=Linux, M=Mac, X=Anything

A = Apache Web Server

M = MySQL , MariaDB

P = PHP

P = Perl

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XAMPP download

v.3.3.0

 **XAMPP** Apache + MariaDB + PHP + Perl

Cos'è XAMPP?

XAMPP è il più popolare ambiente di sviluppo PHP

XAMPP è una distribuzione di Apache completamente gratuita e semplice da installare, contenente MySQL, PHP e Perl. Il pacchetto open source XAMPP è stato creato per essere estremamente facile da installare e utilizzare.



Scarica

Clicca qui per altre versioni

 XAMPP per **Windows**
8.1.12 (PHP 8.1.12)

 XAMPP per **Linux**
8.1.12 (PHP 8.1.12)

 XAMPP per **OS X**
8.1.12 (PHP 8.1.12)

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For Mac users...

LAMPP is installed on a virtual machine

So, we need to:

- Connect the port on our machine (e.g. 8080) to the 80 port on the VM

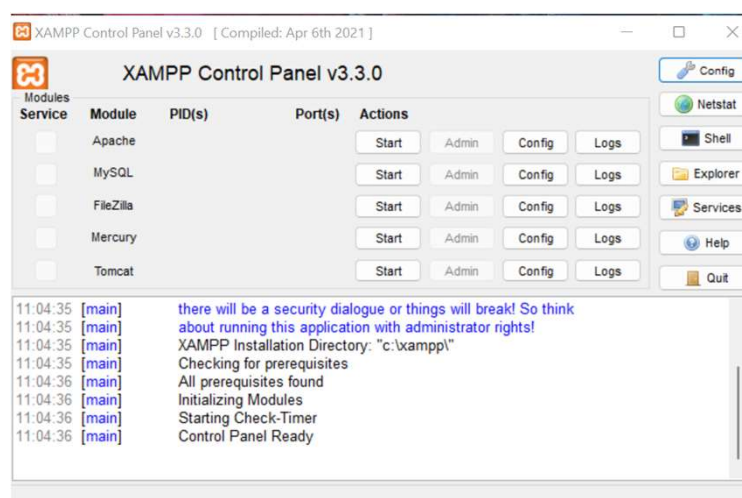
- Access the VM file system from our file system

Mac users: do not use the VM version, but rather use this:

<https://sourceforge.net/projects/xampp/files/XAMPP%20Mac%20OS%20X/7.4.2/>

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XAMPP Control Panel



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XAMPP: Starting an Apache web server

XAMPP Control Panel v3.3.0 [Compiled: Apr 6th 2021]

Service	Module	PID(s)	Port(s)	Actions
☒	Apache	10844 1820	80, 443	Stop Admin Config Logs
☐	MySQL			Start Admin Config Logs
☐	FileZilla			Start Admin Config Logs
☐	Mercury			Start Admin Config Logs
☐	Tomcat			Start Admin Config Logs

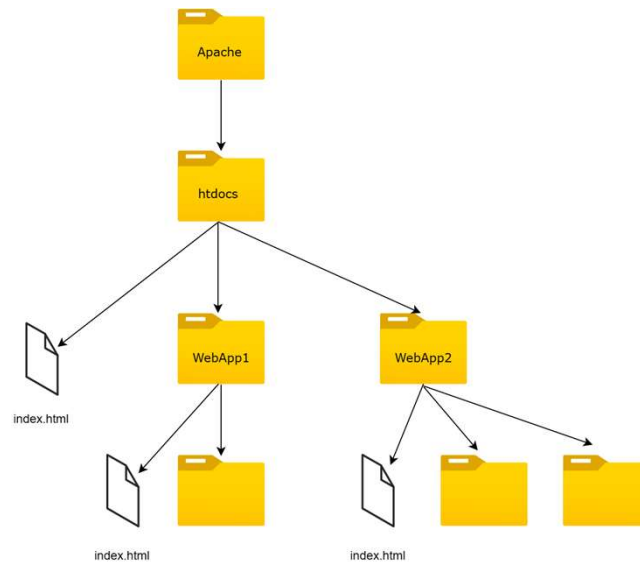
11:04:35 [main] XAMPP Installation Directory: "C:\xampp\"
 11:04:35 [main] Checking for prerequisites
 11:04:36 [main] All prerequisites found
 11:04:36 [main] Initializing Modules
 11:04:36 [main] Starting Check-Timer
 11:04:36 [main] Control Panel Ready
 11:08:06 [Apache] Attempting to start Apache app
 11:08:06 [Apache] Status change detected: running

httpd.conf - Blocco note

```
#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default. See also the <VirtualHost>
# directive.
#
# Change this to Listen on specific IP addresses as shown below to
# prevent Apache from glomming onto all bound IP addresses.
#
#Listen 12.34.56.78:80
Listen 80
```

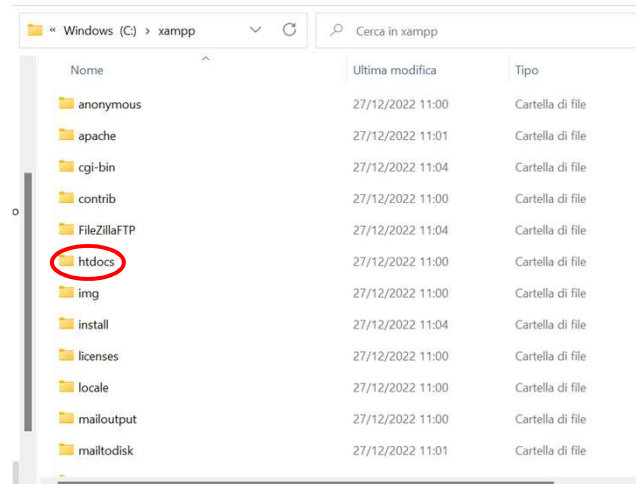
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Directory structure for an Apache web site



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Let's take a look at the XAMPP folders



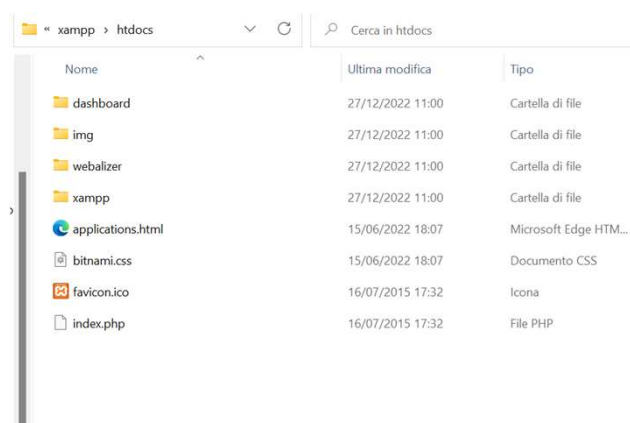
« Windows (C:) > xampp

Cerca in xampp

Nome	Ultima modifica	Tipo
anonymous	27/12/2022 11:00	Cartella di file
apache	27/12/2022 11:01	Cartella di file
cgi-bin	27/12/2022 11:04	Cartella di file
contrib	27/12/2022 11:00	Cartella di file
FileZillaFTP	27/12/2022 11:04	Cartella di file
htdocs	27/12/2022 11:00	Cartella di file
img	27/12/2022 11:00	Cartella di file
install	27/12/2022 11:04	Cartella di file
licenses	27/12/2022 11:00	Cartella di file
locale	27/12/2022 11:00	Cartella di file
mailoutput	27/12/2022 11:00	Cartella di file
mailtodisk	27/12/2022 11:01	Cartella di file

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XAMPP htdocs



« xampp > htdocs

Cerca in htdocs

Nome	Ultima modifica	Tipo
dashboard	27/12/2022 11:00	Cartella di file
img	27/12/2022 11:00	Cartella di file
webalizer	27/12/2022 11:00	Cartella di file
xampp	27/12/2022 11:00	Cartella di file
applications.html	15/06/2022 18:07	Microsoft Edge HTM...
bitnami.css	15/06/2022 18:07	Documento CSS
favicon.ico	16/07/2015 17:32	Icona
index.php	16/07/2015 17:32	File PHP

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